

Data Analysis Report: Telecom

Customer Churn Exploration

1. Executive Summary

This report provides a formal, comprehensive evaluation of the Exploratory Data Analysis (EDA) performed on the telecom customer churn dataset. The primary objective of this analysis is to identify key customer segments, contract characteristics, service attributes, and behavioral trends that correlate with customer attrition.

The analysis reveals an overall **customer churn rate of 26.54%**, meaning approximately one in every four customers leaves the telecom provider. The findings strongly indicate that customer attrition is not random; rather, it is heavily concentrated within specific operational vectors:

- **Contractual Risk:** Month-to-month contracts.
- **Infrastructure Issues:** Fiber optic internet service subscriptions.
- **Customer Maturity:** Short-tenure lifecycles (0–12 months).
- **Value-Added Service Gaps:** Absence of Online Security and Technical Support features.

By strategically targeting these vulnerable dimensions, the business can transition from reactive mitigation to proactive customer retention management, ultimately protecting annual recurring revenue (ARR) and improving Lifetime Value (LTV).

2. Project Context & Objectives

The primary goal of this initiative is to translate raw, customer-level telecom transactional data into actionable business intelligence. The specific objectives guiding the analysis include:

1. **Baseline Metrics:** Establish a precise measurement of the overall customer churn baseline.
2. **Risk Segmentation:** Categorize and isolate high-risk customer groups based on demographic and behavioral traits.
3. **Service and Contract Assessment:** Evaluate the interaction between core product offerings (e.g., Internet Service, Contract Type) and attrition rates.
4. **Support and Security Dependency:** Determine if value-added security and support add-ons act as stabilizing factors for customer retention.
5. **Lifecycle Analysis:** Map the trajectory of churn risk across distinct customer tenure brackets to pinpoint critical exit windows.
6. **Strategic Recommendations:** Formulate data-driven optimization strategies and feature engineering foundations for predictive modeling.

3. Data Processing & Quality Assurance Workflow

To ensure analytical integrity, a structured data engineering and data cleaning workflow was established to eliminate systemic anomalies prior to statistical aggregation.

Key Data Interventions:

- **Structural Inspection:** Profiling data shapes using `df.shape` and `df.info()` to evaluate implicit data types and locate structural discrepancies.
- **Handling Blanks in Financial Data:** The `TotalCharges` column was initially ingested as an object/string due to the presence of blank characters (" "). Converting this field via `pd.to_numeric(..., errors="coerce")` successfully isolated these records as missing values (NaN).
- **Zero-Tenure Imputation:** Downstream filtering confirmed that all missing `TotalCharges` records strictly corresponded to new customers with a `tenure` of 0. Because these individuals had not completed their first billing cycle, their missing financial data was logically imputed with 0 via `df.fillna(0)`, preserving rows that would otherwise be discarded.
- **Categorical Standardization:** The `SeniorCitizen` indicator was transformed from binary integers (1, 0) to human-readable strings ("Yes", "No") via mapping to guarantee unified representation across subsequent categorical plot functions.
- **Deduplication Verification:** The pipeline verified that there were zero complete duplicate rows and zero duplicate `customerID` fields, confirming the dataset uniquely identifies individual accounts.

4. Key Analytical Insights & Data Visualizations

4.1 Overall Churn Distribution

The baseline calculation demonstrates a critical retention challenge. Out of the total customer population analyzed, **26.54%** have separated from the company.

Takeaway: A churn rate exceeding a quarter of the base signifies that acquisition costs (CAC) are being continuously deflated by high attrition, demanding immediate structural intervention.

4.2 Impact of Contract Structures

Contract design proved to be one of the strongest categorical indicators of attrition. When evaluating churn across Contract Types, the results showed an inverse relationship between commitment length and the probability of exit:

- **Month-to-Month Contracts:** Experiencing the highest churn rate by a substantial margin.
- **One-Year & Two-Year Contracts:** Show significantly lower, stable churn profiles.

Strategic Insight: Short-term transactional terms introduce persistent friction points, whereas structured multi-year agreements anchor customer habits and drastically insulate the business against sudden churn.

4.3 The Fiber Optic Paradigm

A major finding emerged within the infrastructure analysis:

- **Fiber Optic Internet Service:** Correlates with the highest churn rates among all internet service tiers, outpacing DSL and non-internet users.

Strategic Insight: While fiber optic networks are typically marketed as premium, high-speed offerings, the elevated churn rates suggest underlying operational friction. This could stem from uncompetitive pricing strategies, mismatched expectations during onboarding, or systemic technical instability and service outages.

4.4 Value-Added Service Vulnerabilities (Security & Technical Support)

Cross-tabulating churn across optional service layers highlighted a stark protective effect provided by specific utility add-ons:

- **Absence of Online Security:** Customers who do not opt into Online Security demonstrate an elevated churn rate compared to protected counterparts.
- **Absence of Technical Support:** Customers without a Technical Support subscription display a parallel, aggressive churn profile.



Strategic Insight: Support and utility features act as retention anchors. Customers who operate without these safety nets are highly vulnerable to technical challenges, making them significantly quicker to abandon the platform.

4.5 Customer Maturity & Tenure Segmentation

To track risk throughout the customer lifecycle, raw continuous tenure data was converted into categorical cohorts using continuous binning ([-1, 12, 24, 48, 72]):

- **0–12 Months Cohort:** Suffers from extreme churn vulnerability. This first-year window accounts for the highest concentration of total churned accounts.
- **25–72 Months Cohorts:** Show progressive stabilization. As an account matures past the 24-month mark, the churn probability falls off drastically.

Strategic Insight: The initial 365 days represents a vulnerable onboarding phase. If a customer encounters technical roadblocks or experiences poor onboarding within this period, their likelihood of early departure peaks.

5. Strategic Recommendations

To reverse these observed trends, the business should shift away from uniform communication strategies and focus on execution across these targeted tactical tracks:

I. Contract Migration Incentives

- **Action:** Launch automated, targeted marketing plays aimed exclusively at Month-to-Month accounts before their standard monthly renewal loop.
- **Tactic:** Offer localized loyalty discounts, temporary service upgrades, or bundled rewards conditional on migrating to a 1-year or 2-year commitment.

II. Audit and Rectify the Fiber Optic Experience

- **Action:** Spin up a cross-functional task force between Network Operations, Quality Assurance, and Product Management to diagnose the root cause of Fiber Optic churn.
- **Tactic:** Investigate the underlying triggers behind fiber optic attrition by cross-referencing account notes for billing disputes, tracking frequent connectivity drops, or reviewing technical support ticket backlogs.

III. Onboarding Optimization for Early-Tenure Cohorts

- **Action:** Build a structured "First-Year Retention Program" focused entirely on accounts in the 0–12 month tenure bracket.
- **Tactic:** Introduce proactive automated outreach cadences at day 30, 90, and 180 to evaluate service satisfaction, clear up setup confusion, and actively guide users through their service deployment.

IV. Service Bundling & Add-on Integration

- **Action:** Re-engineer service package architectures to incentivize the adoption of support features.
- **Tactic:** Repackage high-risk single-product offerings into comprehensive bundles that natively include Online Security and Technical Support. Position these additions as high-value, peace-of-mind add-ons during checkout and within upgrade portals.

6. Conclusion & Next Steps

This exploratory data analysis confirms that customer churn is highly correlated with specific contract types, infrastructure layers, tenure groups, and support architectures. The primary points of vulnerability are **short-tenure, month-to-month accounts** and **fiber optic subscribers who lack value-added support services**.